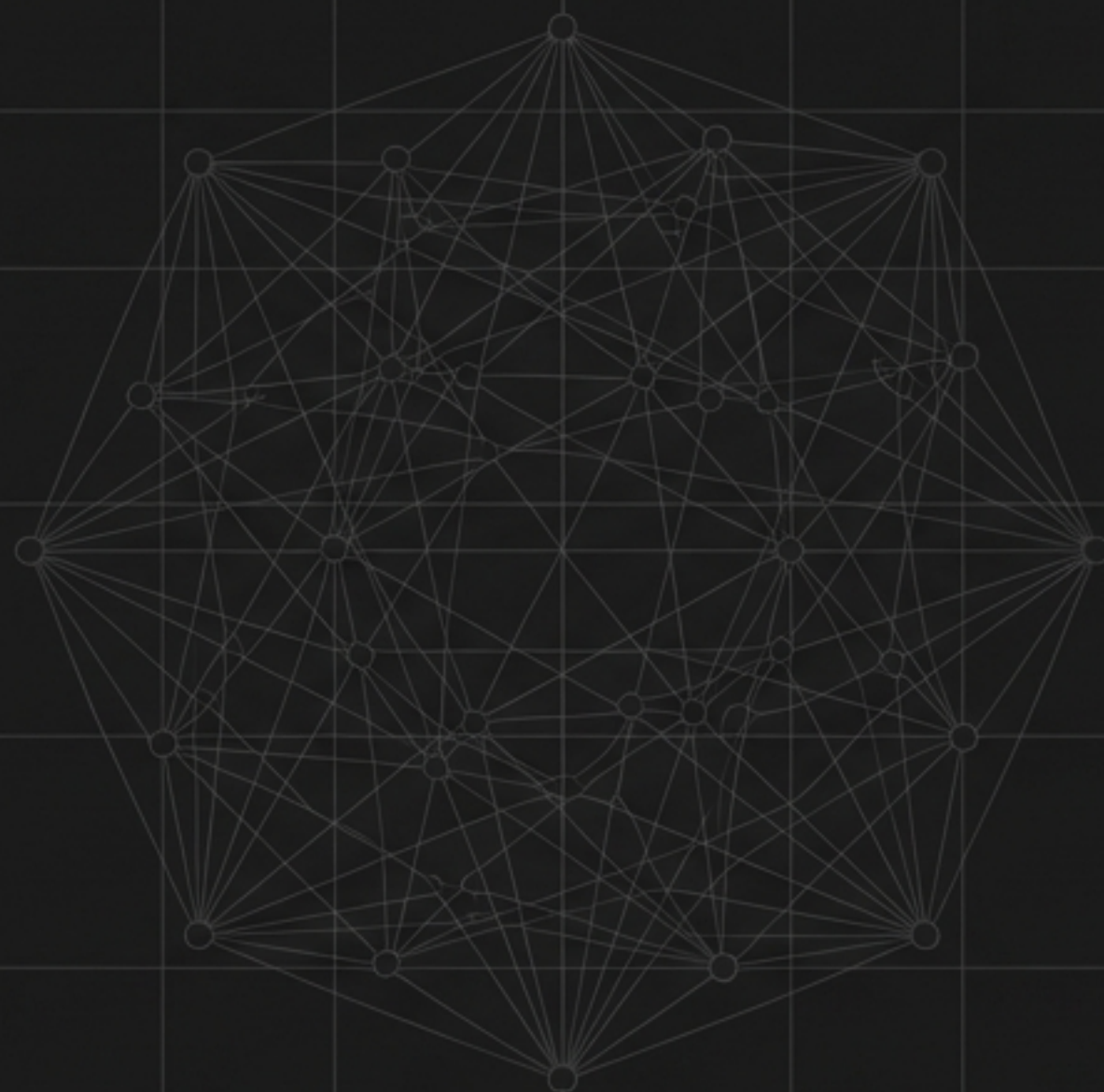


SGraph Send: First MVP User Site

Milestone Record v0.2.15 — End-to-End Zero-Knowledge Encryption



MVP Confirmed: Secure Transfer in 5 Days

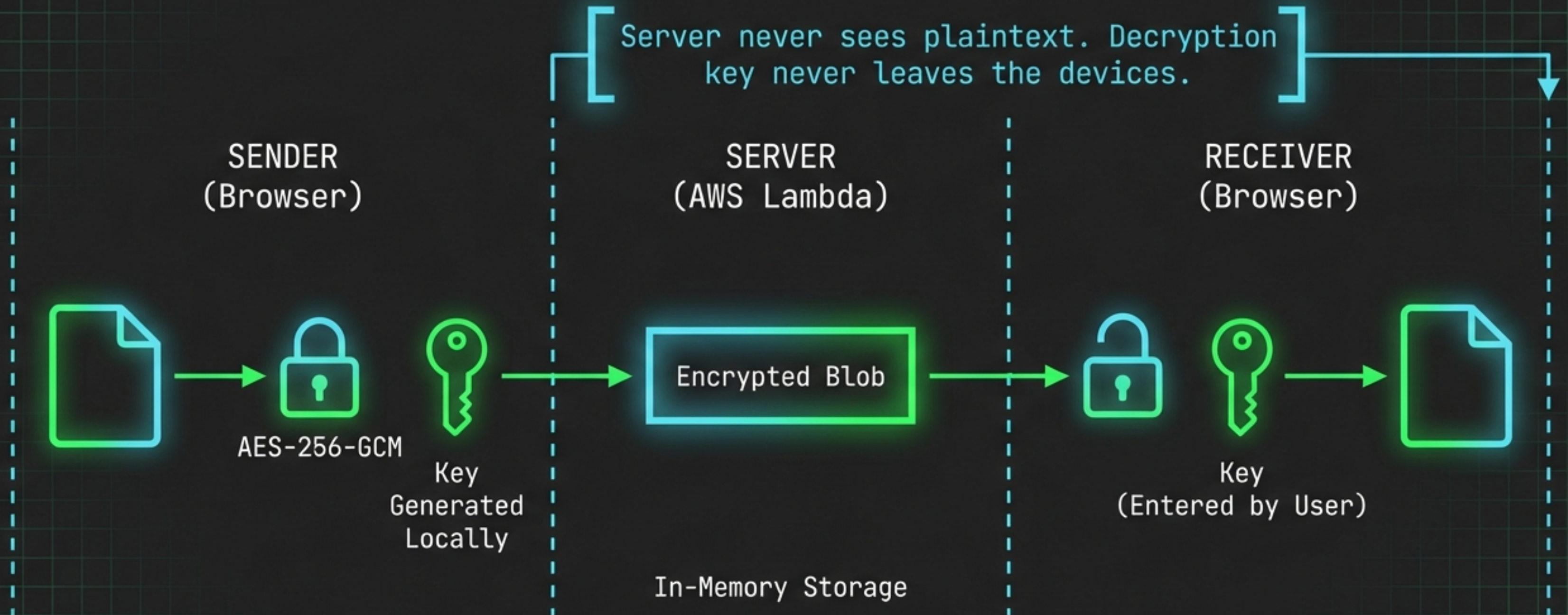
SYSTEM METRICS	
STATUS:	CONFIRMED WORKING
ELAPSED TIME:	5 Days (08 Feb – 12 Feb 2026)
TOTAL COMMITS:	91
TEAM STRUCTURE:	1 Human Lead / 8 AI Roles
A fully functional end-to-end encrypted file transfer system deployed to AWS Lambda.	



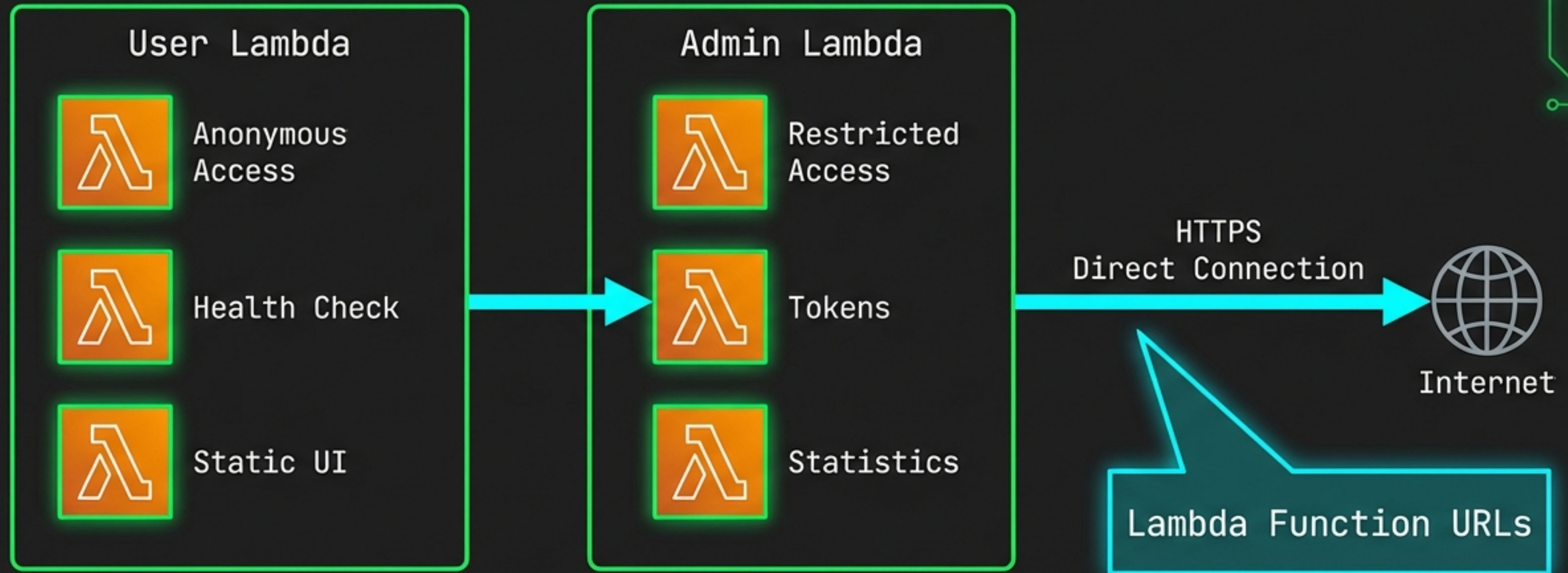
Verification: Manual
Testing by Dinis Cruz

uptime: 120h

The Zero-Knowledge Transfer Cycle

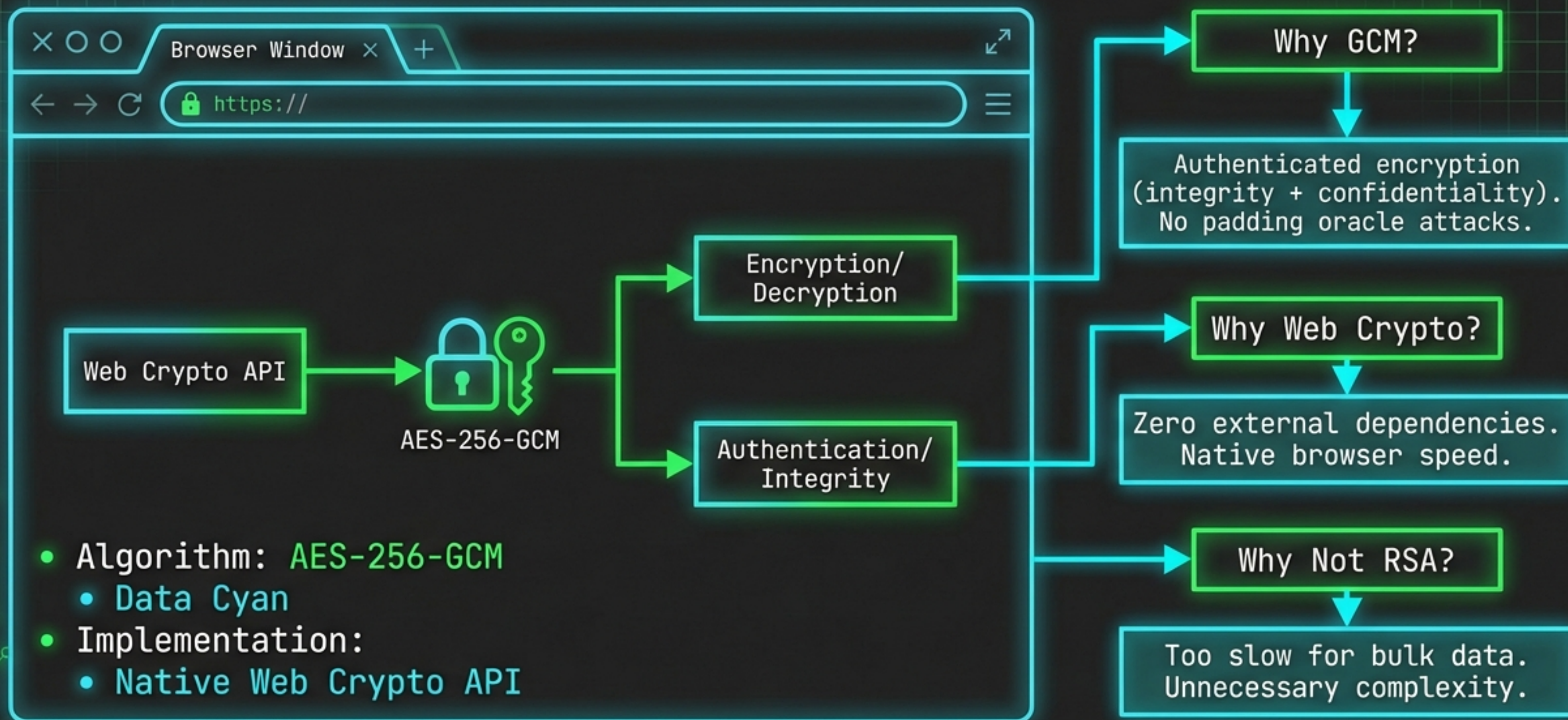


Lean Infrastructure: The Two-Lambda Model



Rejection of API Gateway: Function URLs provide direct endpoints with zero additional infrastructure or cost.

Cryptographic Foundation



Isolated Frontend Development (IFD)

~~Traditional Stack~~

~~node_modules~~

~~webpack~~

~~build steps~~

📁 sgraph-send/

📁 v0.2.10/

📄 index.html

📄 style.css

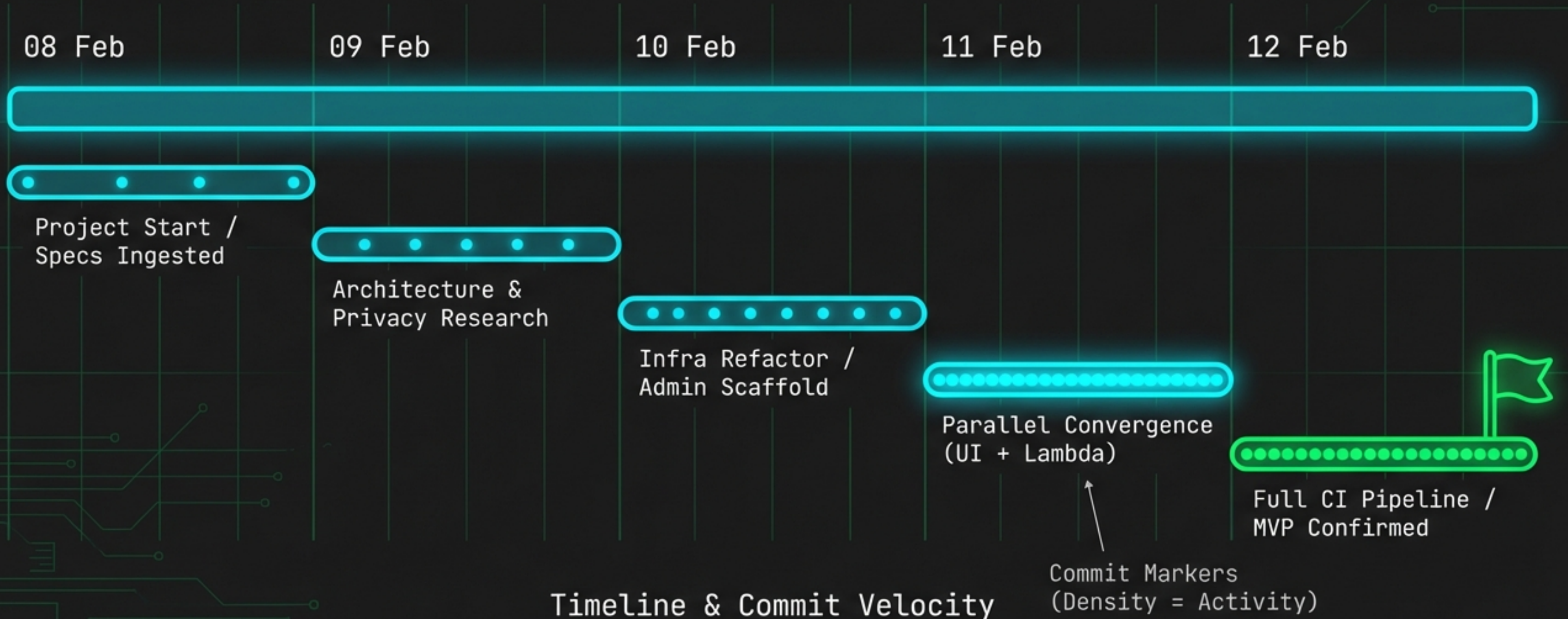
📄 app.js

📁 v0.2.11/

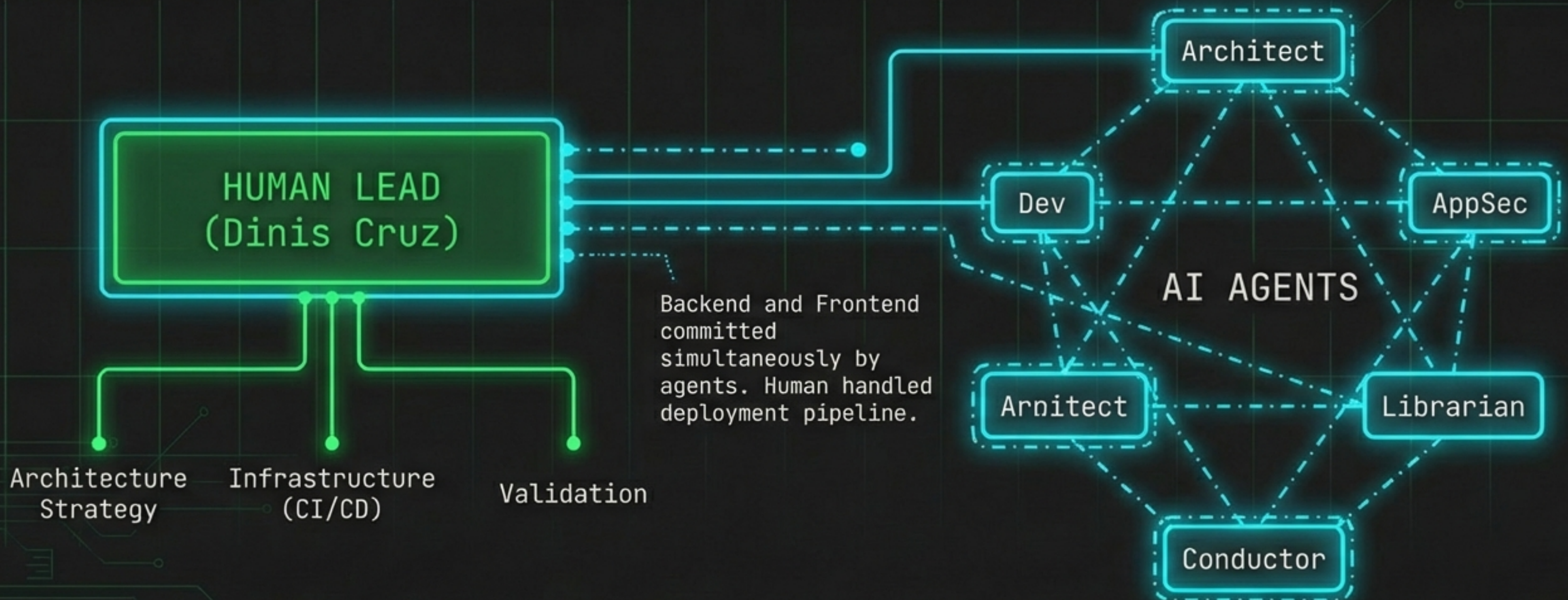
📄 index.html

- Zero build step
- Vanilla JavaScript & Web Components
- Surgical Versioning
(Serve from specific path)
- Instant Deployment

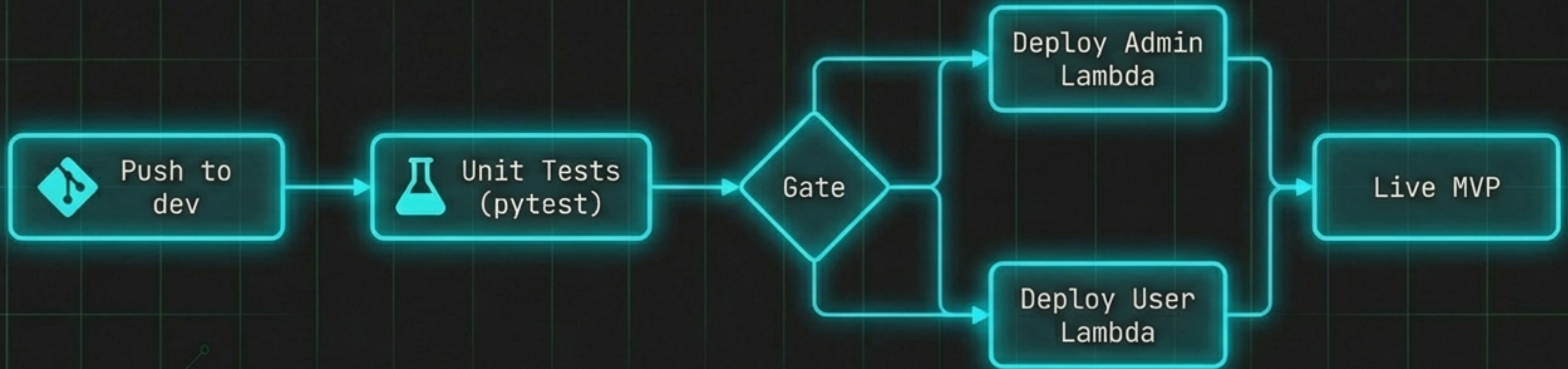
From Zero to MVP in 120 Hours



Agentic Parallel Development



Test-Driven Deployment



“Programmatic creation,
verification, and destruction.”

Pragmatism Over Perfection

AREA	STANDARD SPEC	MVP REALITY
Storage	S3 Bucket	In-Memory (Memory-FS) Speed of prototyping
Network	API Gateway	Function URLs Reduced complexity
Code Schema	Pydantic	Type_Safe Ecosystem control

Deliberate Constraints (MVP Scope)



Ephemeral Storage

Files lost on Lambda recycle. Acceptable for flow validation.



Localhost Issues

Web Crypto API requires Secure Contexts (HTTPS).



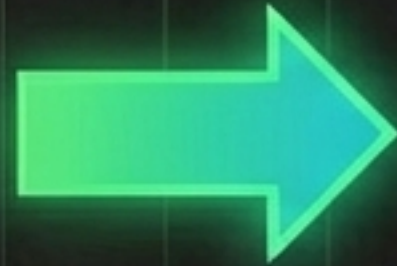
Single Environment

Dev only. QA/Prod deferred.

"I was able to get it working once... which proves the full end to end flow." — Dinis Cruz

Immediate Roadmap (v0.3.0)

Current State
(v0.2.15 (MVP))



Future State (v0.3.0)



Persistence Layer
Switch Memory-FS to S3 Backend



Security Hardening
Audit injection risks (Routes__Transfers)



Refactoring
Hardening schemas with `Type\_Safe`

Validation of Concept & Method

- ✓ **Architecture:** Zero-knowledge works in browser.
- ✓ **Abstraction:** Memory-FS allowed deployment without S3.
- ✓ **Process:** Agentic development + Human oversight = Working Code.

[The core architectural premise is validated.]